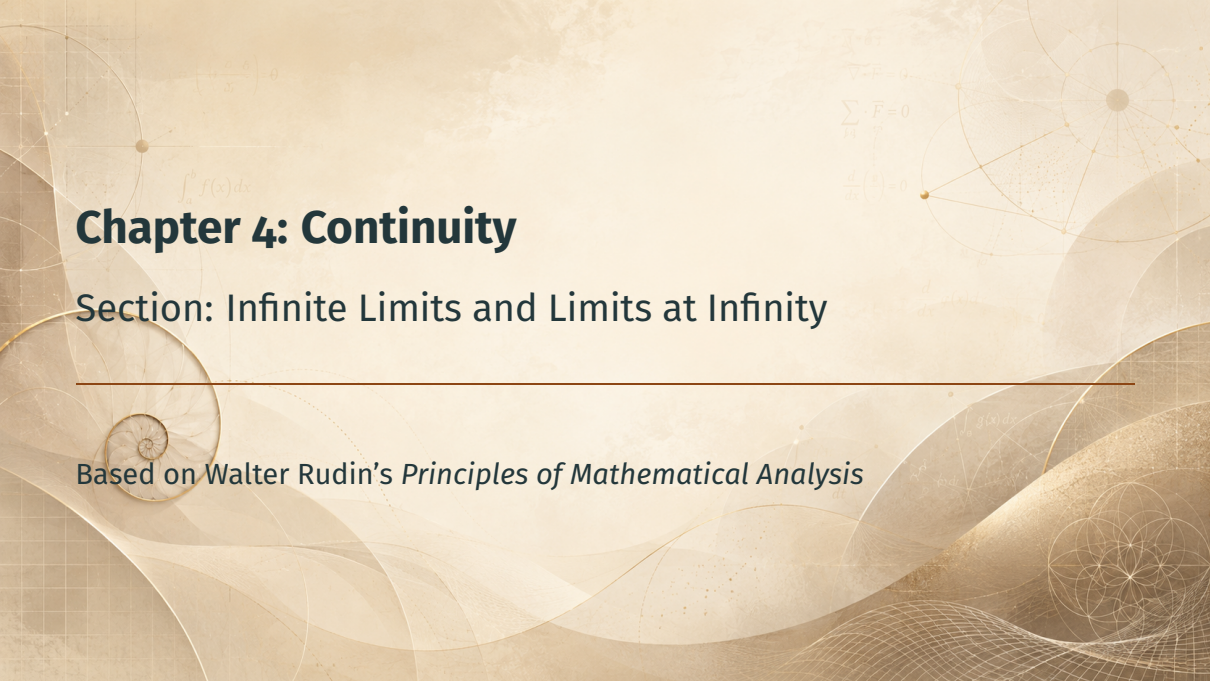


The background is a light beige color with various mathematical and geometric elements. In the top left, there is a partial view of a golden spiral. In the top right, there are faint, overlapping circles and lines, resembling a geometric construction or a celestial map. Scattered throughout are several mathematical formulas in a light, semi-transparent font. These include the vector equation $\vec{\nabla} \cdot \vec{F} = \rho$, the divergence theorem $\sum_{i=1}^n \vec{F} = 0$, the derivative $\frac{d}{dx} \left(\frac{g}{f} \right) = 0$, the integral $\int_a^b f(x) dx$, and the differential equation $\frac{d}{dt} g(x) = 0$.

Chapter 4: Continuity

Section: Infinite Limits and Limits at Infinity

Based on Walter Rudin's *Principles of Mathematical Analysis*

Outline

Motivation

Neighborhoods of Infinity

The Unified Limit

Arithmetic of Limits

Summary

Motivation

Why Extend the Limit?

- So far: $f(t) \rightarrow A$ as $t \rightarrow x$ with $A, x \in \mathbb{R}$

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Why Extend the Limit?

- So far: $f(t) \rightarrow A$ as $t \rightarrow x$ with $A, x \in \mathbb{R}$
- We often want $x = \pm\infty$ or $A = \pm\infty$
- **Key idea:** rephrase Definition 4.1 using *neighborhoods*

$$\nabla \cdot F = 0$$

$$\frac{d}{dx}(e^x) = e^x$$

$$\int_a^b f(x) dx$$

$$e^{i\pi} + 1 = 0$$

Neighborhoods of Infinity

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

Recall: Neighborhood of a Real Number

Neighborhood of a real x

A neighborhood of a real number x is any segment

$$(x - \delta, x + \delta), \quad \delta > 0.$$

Idea: “ t is close to x ” means “ t lies in such a segment.”

Definition 4.32: Neighborhoods of $\pm\infty$

Definition 4.32

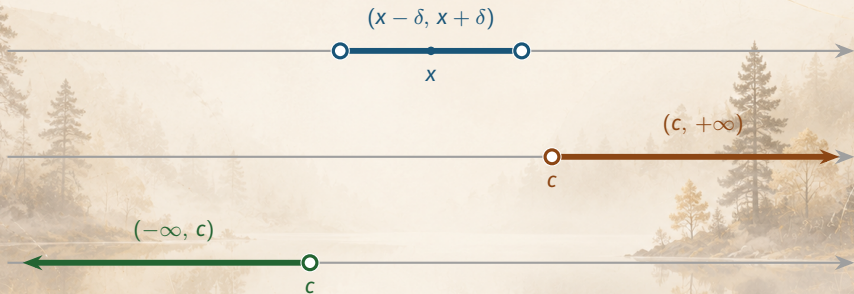
For any real c :

- $(c, +\infty) = \{x : x > c\}$ is a neighborhood of $+\infty$.
- $(-\infty, c) = \{x : x < c\}$ is a neighborhood of $-\infty$.

Idea: “close to $+\infty$ ” means “larger than some cutoff c .”

Picture: Three Kinds of Neighborhood

$$\frac{d}{dx}(e^x) = e^x$$
$$\int_a^b f(x) dx$$



$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

$$e^{i\pi} + 1 = 0$$

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

$$y = \frac{\sin x}{x}$$

The Unified Limit

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Definition 4.33: Limits in the Extended Reals

Definition 4.33

Let f be a real function on $E \subset \mathbb{R}$, and let A, x lie in the extended reals. We write $f(t) \rightarrow A$ as $t \rightarrow x$ provided:

for every neighborhood U of A ,
there is a neighborhood V of x with $V \cap E \neq \emptyset$,
such that $f(t) \in U$ for all $t \in V \cap E$, $t \neq x$.

It Agrees with the Old Definition

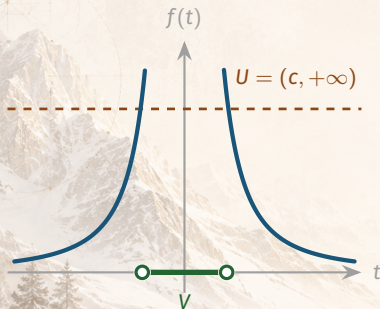
Consistency check

When A and x are both *real*, Definition 4.33 says exactly the same thing as Definition 4.1.

- $U = (A - \varepsilon, A + \varepsilon) \iff |f(t) - A| < \varepsilon$
- $V = (x - \delta, x + \delta) \iff 0 < |t - x| < \delta$

▷ The strict " $0 <$ " comes from the clause $t \neq x$: the limit ignores the value of f at x .

Example: $f(t) \rightarrow +\infty$ as $t \rightarrow 0$



$$f(t) = 1/t^2$$

Given any cutoff c , choose δ with $1/\delta^2 > c$. Then

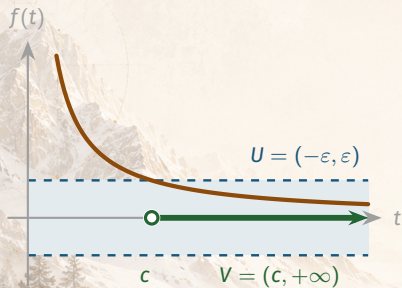
$$0 < |t| < \delta \implies f(t) > c.$$

So $f(t) \rightarrow +\infty$ as $t \rightarrow 0$.

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Example: $f(t) \rightarrow 0$ as $t \rightarrow +\infty$



$f(t) = 1/t$ on $(0, \infty)$

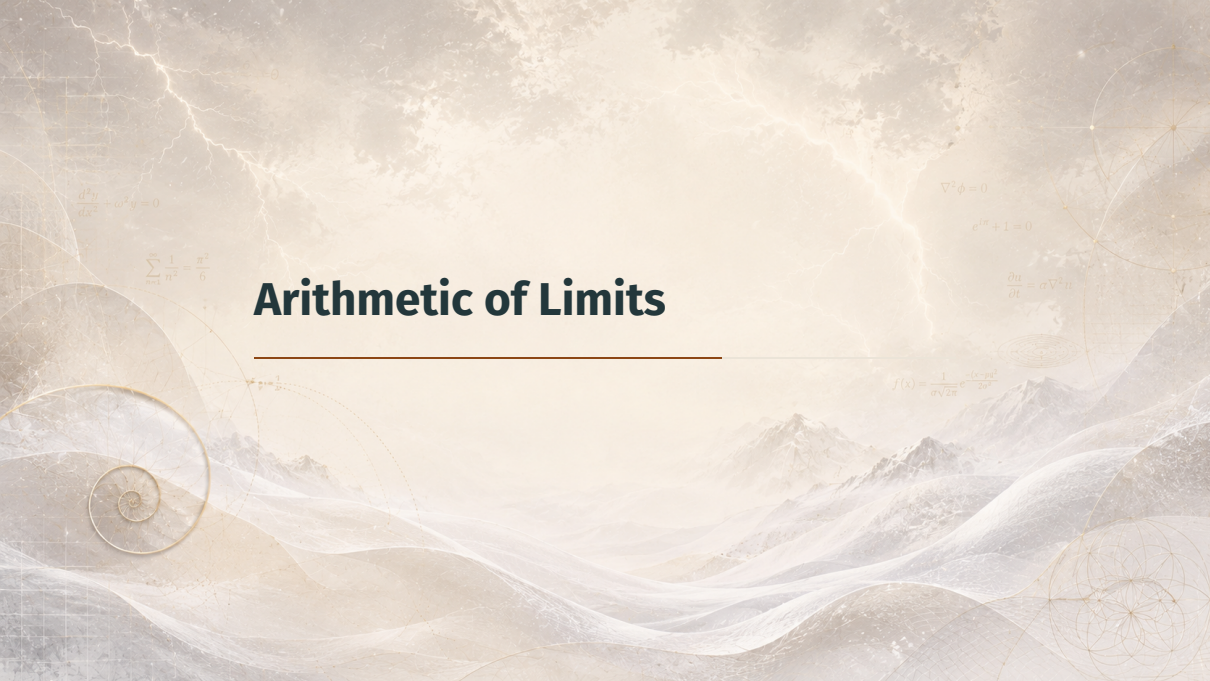
Given any $\varepsilon > 0$, take the cutoff $c = 1/\varepsilon$. Then

$$t > c \implies |f(t)| < \varepsilon.$$

So $f(t) \rightarrow 0$ as $t \rightarrow +\infty$.

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$



$$\frac{d^2 y}{dx^2} + \omega^2 y = 0$$

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

Arithmetic of Limits

$$\nabla^2 \phi = 0$$

$$e^{im} + 1 = 0$$

$$\frac{\partial u}{\partial t} = \sigma \nabla^2 u$$

$$f(x) = \frac{1}{\sigma \sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

$$\lim_{x \rightarrow \infty} \frac{1}{x^2}$$

Theorem 4.34: Limit Arithmetic

Theorem 4.34

Let f, g be defined on $E \subset \mathbb{R}$, and suppose

$$f(t) \rightarrow A, \quad g(t) \rightarrow B \quad \text{as } t \rightarrow x.$$

Then, *provided the right-hand sides are defined:*

(a) the limit is **unique**: $f(t) \rightarrow A'$ implies $A' = A$.

Theorem 4.34: Limit Arithmetic

Theorem 4.34 (continued)

Provided the right-hand sides are defined in the extended reals:

$$(b) (f + g)(t) \rightarrow A + B$$

$$(c) (fg)(t) \rightarrow AB$$

$$(d) (f/g)(t) \rightarrow A/B$$

The Forbidden Forms

Note (see Definition 1.23)

The following are **not defined** in the extended reals:

$$\infty - \infty, \quad 0 \cdot \infty, \quad \frac{\infty}{\infty}, \quad \frac{A}{0}.$$

Theorem 4.34 applies *only* when the right-hand side avoids these.

The background is a complex collage. It features a misty landscape with a lake, mountains, and cherry blossom trees. Overlaid on this are various mathematical diagrams: a coordinate plane with a point labeled 'a', a complex plane with points labeled 'z' and 'z-bar', a golden spiral, and several geometric constructions involving circles and lines. The overall color palette is warm, with soft pinks, oranges, and yellows.
$$e^{i\pi} + 1 = 0$$

$$\phi = \frac{1 + \sqrt{5}}{2}$$

Summary

$$\nabla^2 \psi + k^2 \psi = 0$$

$$\frac{d}{dx}(\sin x) = \cos x$$

Summary

Key takeaways:

- Neighborhoods of $\pm\infty$: rays $(c, +\infty)$ and $(-\infty, c)$

$$\nabla^2 \psi + k^2 \psi = 0$$

$$\frac{d}{dx}(\sin x) = \cos x$$

Summary

Key takeaways:

- Neighborhoods of $\pm\infty$: rays $(c, +\infty)$ and $(-\infty, c)$
- One definition (4.33) covers all finite and infinite limits

$$\nabla^2 \psi + k^2 \psi = 0$$

$$\frac{d}{dx}(\sin x) = \cos x$$

Summary

Key takeaways:

- Neighborhoods of $\pm\infty$: rays $(c, +\infty)$ and $(-\infty, c)$
- One definition (4.33) covers all finite and infinite limits
- Arithmetic (4.34) carries over — except the **undefined forms**

Coming up next: Chapter 5 — Differentiation.